

Supplementary Proofs for Cancellation-Preserving Optimistic Online Learning under Delayed Feedback

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This document contains the proof targets and complete proofs for the results in the main paper. Algorithm descriptions, experiments, and duplicate exposition are omitted.

Notation used in the proofs

Let $\mathcal{X} \subseteq \mathbb{R}^d$ be the feasible set, let $u \in \mathcal{X}$ be a static comparator, and let $\mathcal{O}_t$, $\mathcal{U}_t$, and $\mathcal{V}_t = \mathcal{U}_t \cup \{t\}$ denote the observed, unobserved, and outstanding index sets. Write

$$\begin{aligned} n_t &= |\mathcal{U}_t|, & k_t &= |\mathcal{V}_t| = n_t + 1, \\ e_t^g &= g_t - \hat{g}_t, & E_t^g &= \sum_{s \in \mathcal{U}_t} e_s^g. \end{aligned}$$

Let $P_t \succeq \mu I$, $A_0 = \lambda \mu I$, and $A_t = A_0 + \sum_{s \leq t} P_s$. Define

$$\begin{aligned} h_T &= \sum_{t \leq T} \frac{1}{t}, & M_T &= \sum_{t \leq T} \frac{n_t}{t}, \\ \sigma_T &= \max_{t \leq T} n_t, & \mathcal{D}_T &= \sum_{t \leq T} n_t. \end{aligned}$$

For the safeguarded update, set

$$b_t^g = \sum_{s \in \mathcal{V}_t} g_s, \quad h_t^g = \sum_{s \in \mathcal{V}_t} \hat{g}_s, \quad c_{t-1} = \sum_{s < t} P_s x_s.$$

The projection and implementable update are

$$q_t = \arg \min_{\|q\|_2 \leq k_t G} \frac{1}{2} \|q - h_t^g\|_{A_t^{-1}}^2, \tag{1}$$

$$x_t \in \arg \min_{x \in \mathcal{X}} \left\{ \frac{1}{2} x^\top A_{t-1} x + \left\langle \sum_{s \in \mathcal{O}_t} g_s + q_t - c_{t-1}, x \right\rangle \right\}. \tag{2}$$

The equivalent centered form used in the proofs is

$$x_t \in \arg \min_{x \in \mathcal{X}} \left\{ \frac{\lambda \mu}{2} \|x\|_2^2 + \frac{1}{2} \sum_{s < t} \|x - x_s\|_{P_s}^2 + \left\langle \sum_{s \in \mathcal{O}_t} g_s + q_t, x \right\rangle \right\}. \quad (3)$$

The delay envelope is

$$\mathfrak{M}_T(\bar{\sigma}, D) = \max \left\{ \sum_{t=1}^T \frac{n_t}{t} : (n_t) \text{ is } (\bar{\sigma}, D)\text{-admissible} \right\}, \quad (4)$$

and the scalar quadratic class and minimax value are

$$\mathcal{X} = [-\varepsilon/\mu, \varepsilon/\mu], \quad f_t(x) = \frac{\mu}{2} x^2 + a_t x, \quad |a_t| \leq \varepsilon, \quad (5)$$

$$\mathcal{R}_T^*(\bar{\sigma}, D) = \inf_{\mathcal{A}} \sup_{I \in \mathcal{I}_T(\bar{\sigma}, D)} \mathbb{E}_{\mathcal{A}} \left[\sum_{t=1}^T (f_t(x_t) - f_t(u^*)) \right]. \quad (6)$$

Proof targets

The following statements are included only to make the proofs self-contained; their motivation and interpretation are given in the main paper.

Assumption 1 (Domain and computation). $\mathcal{X} \subseteq \mathbb{R}^d$ is nonempty, closed and convex with $0 \in \mathcal{X}$; it is compact when $\lambda = 0$. Every stated convex program is solved exactly, and the least-Euclidean-norm minimizer is used to break first-round ties.

Assumption 2 (Curvature and feedback). A known $\mu > 0$ is available. Before round t , a finite symmetric $P_t \succeq \mu I$ is revealed and satisfies

$$f_t(u) \geq f_t(x) + \langle \nabla f_t(x), u - x \rangle + \frac{1}{2} \|u - x\|_{P_t}^2 \quad (7)$$

for all $x, u \in \mathcal{X}$. The only feedback is $g_t = \nabla f_t(x_t)$, delivered at the end of round τ_t , and a known $G > 0$ satisfies $\|g_t\|_2 \leq G$ along the realized trajectory.

Assumption 3 (Timing and hints). Feedback may arrive out of order, after the horizon, or never; each item is indexed and processed once. The learner is not given T , σ_T , $\mathcal{D}_T$, future arrival times, or loss values. Before choosing x_t , it receives a finite hint $\hat{g}_t$, which is then frozen and otherwise unrestricted in magnitude and accuracy.

Lemma 1 (Weighted Cauchy Schwarz, Young, and inverse-norm control). *Let $M \succ 0$. For any vectors v, w ,*

$$|v^\top w| \leq \|v\|_{M^{-1}} \|w\|_M. \quad (8)$$

For any $\eta > 0$,

$$2v^\top M w \leq \eta \|v\|_M^2 + \eta^{-1} \|w\|_M^2, \quad (9)$$

and therefore

$$\|v + w\|_M^2 \leq (1 + \eta) \|v\|_M^2 + (1 + \eta^{-1}) \|w\|_M^2. \quad (10)$$

If in addition $M \succeq \alpha I$ with $\alpha > 0$, then

$$\|v\|_{M^{-1}}^2 \leq \|v\|_2^2 / \alpha.$$

The same argument yields the weighted triangle inequality $\|v + w\|_M \leq \|v\|_M + \|w\|_M$.

Lemma 2 (Existence, uniqueness, and first-order optimality). *Let $A \succ 0$, $c \in \mathbb{R}^d$, and let $\mathcal{X}$ be nonempty, closed and convex. The quadratic*

$$Q(x) = \frac{1}{2}x^\top Ax + c^\top x$$

has a unique minimizer x_ on $\mathcal{X}$. A point $x_* \in \mathcal{X}$ is that minimizer if and only if*

$$\langle Ax_* + c, z - x_* \rangle \geq 0 \quad \text{for every } z \in \mathcal{X},$$

in which case $Q(z) \geq Q(x_) + \frac{1}{2}\|z - x_*\|_A^2$.*

Lemma 3 (Aggregate projection does not inflate the residual). *Let $M \succ 0$, let $\mathcal{C}$ be nonempty, closed and convex, and let*

$$p = \arg \min_{z \in \mathcal{C}} \frac{1}{2}\|z - h\|_M^2.$$

Then for every $b \in \mathcal{C}$,

$$\|p - b\|_M^2 \leq \|h - b\|_M^2 - \|h - p\|_M^2 \leq \|h - b\|_M^2. \quad (11)$$

If $\mathcal{C} = \{z : \|z\|_M \leq r\}$, then $\|p - b\|_M \leq 2r$ and p is the radial clip of h onto that ball.

Lemma 4 (Displacement under a linear perturbation). *Let Q be as in Lemma 2 and $r \in \mathbb{R}^d$. If*

$$x = \arg \min_{z \in \mathcal{X}} \{Q(z) - r^\top z\}, \quad z_* = \arg \min_{z \in \mathcal{X}} Q(z),$$

then

$$\|x - z_*\|_A \leq \|r\|_{A^{-1}}, \quad 0 \leq Q(x) - Q(z_*) \leq \frac{1}{2}\|r\|_{A^{-1}}^2.$$

Lemma 5 (Quadratic and linear control of the per-round gap). *Let $f_t(x) = \frac{1}{2}x^\top H_t x + a_t^\top x$ with $H_t \succeq 0$, let F_t have curvature $A_t \succ 0$, and suppose F_{t-1} attains its minimum m_{t-1} on $\mathcal{X}$. If $x_t \in \arg \min_{x \in \mathcal{X}} \{F_t(x) - r_t^\top x\}$, then with*

$$g_t = H_t x_t + a_t,$$

$$\Delta_t \leq \min \left\{ \frac{1}{2}\|r_t\|_{A_t^{-1}}^2, \|g_t\|_{A_t^{-1}}\|r_t\|_{A_t^{-1}} \right\}. \quad (12)$$

Lemma 6 (Regret identity). *Let $F_t = R_0 + \sum_{s \leq t} f_s$ with $m_0 = \min R_0 = 0$. For every $u \in \mathcal{X}$,*

$$R_T(u) = R_0(u) + \sum_{t=1}^T \Delta_t - [F_T(u) - m_T], \quad (13)$$

and in particular $R_T(u) \leq R_0(u) + \sum_{t=1}^T \Delta_t$.

Lemma 7 (Surrogate transfer and realizable perturbation). *Under the standing assumptions above,*

$$R_T(u) \leq \sum_{t=1}^T [\ell_t(x_t) - \ell_t(u)] \quad \text{for every } u \in \mathcal{X}.$$

Writing $b_t^g = \sum_{s \in \mathcal{V}_t} g_s$ and $r_t^g = b_t^g - q_t$, the played point x_t minimizes $F_t(x) - \langle r_t^g, x \rangle$. The surrogate F_t has curvature A_t , and $\nabla \ell_t(x_t) = g_t$.

Lemma 8 (Two-sided safety of the anisotropic projection). *Write*

$$\rho_t^g = \|e_t^g + E_t^g\|_{A_t^{-1}}.$$

Then

$$\|r_t^g\|_{A_t^{-1}} \leq \rho_t^g, \quad \|r_t^g\|_2 \leq 2k_t G, \quad \|g_t\|_{A_t^{-1}} \leq \frac{G}{\sqrt{\mu(t+\lambda)}}.$$

Theorem 1 (Local geometry and a robust safeguard, simultaneously). *Under the standing assumptions above, the safeguarded update satisfies, for every $u \in \mathcal{X}$,*

$$R_T(u) \leq \frac{\lambda\mu}{2} \|u\|_2^2 + \sum_{t=1}^T \min\{\alpha_t, \beta_t, \gamma_t\}, \quad (14)$$

where

$$\alpha_t = \frac{(\rho_t^g)^2}{2}, \quad \beta_t = \frac{G\rho_t^g}{\sqrt{\mu(t+\lambda)}}, \quad \gamma_t = \frac{2G^2 k_t}{\mu(t+\lambda)}.$$

The same trajectory therefore obeys

$$R_T(u) \leq \frac{\lambda\mu}{2} \|u\|_2^2 + \min\{Q_T^{\text{res}}, S_{T,\lambda}\}, \quad (15)$$

with residual complexity

$$Q_T^{\text{res}} = \sum_t \|e_t^g\|_{A_t^{-1}}^2 + \sum_t \|E_t^g\|_{A_t^{-1}}^2$$

and safeguard

$$S_{T,\lambda} = \frac{2G^2}{\mu} (h_{T,\lambda} + M_{T,\lambda}), \quad h_{T,\lambda} = \sum_t \frac{1}{t+\lambda}, \quad M_{T,\lambda} = \sum_t \frac{n_t}{t+\lambda}.$$

Corollary 1 (Perfect hints, zero hints, and cancellation). *If $\lambda = 0$ and every gradient hint is exact, then $R_T(u) \leq 0$ for all u . If every hint is zero, the factor 2 in front of the safeguard of (15) improves to 1. On any round with $E_t^g = 0$, the outstanding-residual square in (15) contributes nothing.*

Lemma 9 (Counting identity and two crude envelopes). *On any legal schedule,*

$$\mathcal{D}_T = \sum_{t=1}^T n_t = \sum_{s=1}^T \min\{\tau_s - s, T - s\},$$

with the convention $\min\{\infty, T - s\} = T - s$. For every $\lambda \geq 0$,

$$M_{T,\lambda} \leq \min\{\sigma_T h_{T,\lambda}, 2\sqrt{\mathcal{D}_T}\}, \quad h_{T,\lambda} \leq 1 + \log T.$$

Theorem 2 (Closed form and attainability). *Fix $T \geq 1$ and integers $0 \leq \bar{\sigma} \leq T - 1$, $D \geq 0$. Define*

$$w_t = \min\{\bar{\sigma}, t - 1\}, \quad C_j = \sum_{t=1}^j w_t, \quad C_0 = 0, \quad \delta = \min\{D, C_T\}.$$

Set

$$m = \max\{j \in \{0, \dots, T\} : C_j \leq \delta\}.$$

Then $m \geq 1$. If $m < T$, write $b = \delta - C_m$, so $0 \leq b < w_{m+1}$. The envelope (4) equals

$$\mathfrak{M}_T(\bar{\sigma}, D) = \begin{cases} \sum_{t=1}^T w_t/t, & m = T, \\ \sum_{t=1}^m w_t/t + b/(m+1), & m < T. \end{cases} \quad (16)$$

The value is attained by a legal feedback schedule; it is not a relaxation that ignores timing constraints.

Lemma 10 (A peak consumes a triangular budget). *Any legal schedule that is missing r past observations at a single round satisfies*

$$\mathcal{D}_T \geq \frac{r(r+1)}{2}.$$

Consequently every schedule with total budget D has

$$\max_t n_t \leq \left\lfloor \frac{\sqrt{1+8D}-1}{2} \right\rfloor.$$

Theorem 3 (Explicit majorants). *If $1 \leq \bar{\sigma} \leq T - 1$, then*

$$\mathfrak{M}_T(\bar{\sigma}, D) \leq \min\{2\sqrt{D}, s[1 + \log(T/s)], s[1 + \log(1 + D/s^2)]\}, \quad (17)$$

where $s = \bar{\sigma}$. If $\bar{\sigma} = 0$ or $D = 0$, then $\mathfrak{M}_T = 0$.

Theorem 4 (Stronger simultaneous guarantee, no extra information). *Take $\lambda = 0$. On the realized schedule of the safeguarded update,*

$$R_T(u) \leq \min\left\{Q_T^{\text{res}}, \frac{2G^2}{\mu} [h_T + \mathfrak{M}_T(\sigma_T, \mathcal{D}_T)]\right\}. \quad (18)$$

The same replacement applies to the known-quadratic specialization recorded as Theorem 9 in Appendix B, with leading constant $4L^2$ in place of $2G^2/\mu$. Neither algorithm is told T , σ_T , $\mathcal{D}_T$, or which branch is smaller.

Corollary 2 (A growing gap between displayed certificates). *Let $T \rightarrow \infty$ through the integers and set $s_T = \lfloor T/(\log T)^2 \rfloor$ together with $D_T = C_T(s_T) = s_T T - s_T(s_T + 1)/2$. Then $\min\{s_T h_T, \sqrt{D_T}\} = \Theta(T/\log T)$, whereas*

$$h_T + \mathfrak{M}_T(s_T, D_T) = \Theta(T \log \log T / (\log T)^2). \quad (19)$$

The two displayed upper-bound certificates therefore separate by a factor $\Theta(\log T / \log \log T)$. The comparison is not a pathwise statement about realized regret, and it does not assert that every previous analysis is unable to recover the tighter envelope upon re-examination.

Lemma 11 (Posterior computations for the two-point mixture). *Let p be uniform on $[0, 1]$ and $\theta = 2p - 1$. Conditionally on p , the signs Z_j are independent with $\mathbb{P}(Z_j = 1 \mid p) = p$. After n signs, of which K_n are positive,*

$$m_n := \mathbb{E}[\theta \mid Z_1, \dots, Z_n] = \frac{2K_n - n}{n + 2}, \quad \mathbb{E}[m_n^2] = \frac{n}{3(n + 2)}. \quad (20)$$

Moreover $\mathbb{E}[\theta^2] = 1/3$, and $\mathbb{E}[Z_i Z_j] = 1/3$ whenever $i \neq j$. The count K_n is itself uniform on $\{0, \dots, n\}$.

Theorem 5 (Exact Bayes risk for arbitrary block lengths). *Partition $[T]$ into N contiguous blocks of prescribed lengths $w_1, \dots, w_N$ with*

$$\sum_{j=1}^N w_j = T.$$

Draw signs as in Lemma 11 and set $a_t = \varepsilon Z_j$ throughout block j , with every observation in the block released only at the block's end. Even if the learner is told the block lengths and the prior, the minimal expected static regret on this instance distribution equals

$$\mathcal{B}(w_1, \dots, w_N) = \frac{\varepsilon^2}{3\mu} \left[\frac{\sum_{j=1}^N w_j^2}{T} + \sum_{j=1}^N \frac{w_j}{j+1} \right]. \quad (21)$$

The identity is the Bayes risk of this prior, not a claim that the numerical prefactor is the exact minimax constant against every adversarial sequence.

Corollary 3 (A genuinely multi-round lower bound, and a matching family). *If $T = N\ell$ and every block has length $\ell \geq 1$, then*

$$\mathcal{B} = \frac{\varepsilon^2 \ell}{3\mu} h_{N+1},$$

with peak $\ell - 1$ and total backlog $N\ell(\ell - 1)/2$. For every algorithm there is a deterministic block-sign sequence on which the expected regret is at least this Bayes value. On the same schedules, the known-quadratic specialization with zero hints and $\lambda = 0$ has a Theorem 9 certificate of at most $(2\varepsilon^2 \ell / \mu) h_N$. Along this coherent-residual family the certificate and the information lower bound therefore differ by at most a factor of 6.

Lemma 12 (A budget-feasible block family). *Let $T \geq 2$, $\bar{\sigma} \geq 1$ and $D \geq 1$. Set*

$$\begin{aligned} k &= \min\{\bar{\sigma}, T - 1, \\ &\quad \lfloor (\sqrt{1 + 8D} - 1)/2 \rfloor\}, \\ \ell &= k + 1, \quad C = k(k + 1)/2, \\ N &= \min\{\lfloor T/\ell \rfloor, \lfloor D/C \rfloor\}. \end{aligned}$$

Then $k \geq 1$, $N \geq 1$, and

$$\mathfrak{M}_T(\bar{\sigma}, D) \leq 3\ell h_N.$$

The schedule consisting of N leading blocks of length ℓ , followed by singleton blocks, respects both original budgets.

Theorem 6 (Constant-factor minimax characterization). *For the class defined in (5) and (6), and integers $T \geq 1$, $0 \leq \bar{\sigma} \leq T - 1$, $D \geq 0$,*

$$\frac{\varepsilon^2}{36\mu} [h_T + \mathfrak{M}_T(\bar{\sigma}, D)] \leq \mathcal{R}_T^*(\bar{\sigma}, D) \leq \frac{2\varepsilon^2}{\mu} [h_T + \mathfrak{M}_T(\bar{\sigma}, D)]. \quad (22)$$

The two sides refer to the same class. The upper-bound algorithm does not need the budgets in advance; the lower bound permits algorithms that know the budgets and the schedule used in the construction.

Corollary 4 (Dimension-free characterization for a known matrix). *Fix $d \geq 1$, $H \succ 0$ and $L > 0$, and let $\mathcal{X}$ be a closed convex set containing the entire H -ball $\{x : \|x\|_H \leq L\}$. For zero hints and losses*

$$f_t(x) = \frac{1}{2}x^\top Hx + a_t^\top x, \quad \|a_t\|_{H^{-1}} \leq L,$$

the minimax regret $\mathcal{R}_{T,H,L,\mathcal{X}}^$ under the same two delay budgets satisfies*

$$\frac{L^2}{36} [h_T + \mathfrak{M}_T(\bar{\sigma}, D)] \leq \mathcal{R}_{T,H,L,\mathcal{X}}^*(\bar{\sigma}, D) \leq 2L^2 [h_T + \mathfrak{M}_T(\bar{\sigma}, D)]. \quad (23)$$

The lower bound requires the domain to contain the ball: the singleton $\mathcal{X} = \{0\}$ would make regret identically zero.

Corollary 5 (Normalized strongly convex first-order class). *Fix $d \geq 1$, $\mu, G > 0$ and*

$$\mathcal{X} = \{x \in \mathbb{R}^d : \|x\|_2 \leq G/(2\mu)\}.$$

Let $\mathcal{F}_{\mu,G}$ be the class of differentiable μ -strongly convex losses on this domain with $\|\nabla f_t(x)\|_2 \leq G$ at every feasible x . With zero hints and delayed gradients only, the class minimax value satisfies

$$\frac{G^2}{144\mu} [h_T + \mathfrak{M}_T(\bar{\sigma}, D)] \leq \mathcal{R}_{T,\mu,G}^{\text{sc},*}(\bar{\sigma}, D) \leq \frac{G^2}{\mu} [h_T + \mathfrak{M}_T(\bar{\sigma}, D)]. \quad (24)$$

A Preliminaries

Proof of Lemma 1. If $w = 0$, both sides of (8) vanish. If $w \neq 0$, positivity of M gives, for every real s ,

$$0 \leq (M^{-1}v - sw)^\top M(M^{-1}v - sw) = \|v\|_{M^{-1}}^2 - 2s v^\top w + s^2 \|w\|_M^2.$$

Substituting $s = (v^\top w)/\|w\|_M^2$ yields $0 \leq \|v\|_{M^{-1}}^2 - (v^\top w)^2/\|w\|_M^2$. Multiplying by the positive number $\|w\|_M^2$ and taking the nonnegative square root produces (8).

For (9), expand $0 \leq \|\sqrt{\eta}v - \eta^{-1/2}w\|_M^2 = \eta\|v\|_M^2 - 2v^\top Mw + \eta^{-1}\|w\|_M^2$ and rearrange. Expanding $\|v + w\|_M^2$ and substituting (9) gives (10).

Replacing the first vector in (8) by Mv produces $|v^\top Mw| = |(Mv)^\top w| \leq \|Mv\|_{M^{-1}}\|w\|_M = \|v\|_M\|w\|_M$, using $(Mv)^\top M^{-1}(Mv) = v^\top Mw$ in the final equality. Hence $\|v + w\|_M^2 \leq (\|v\|_M + \|w\|_M)^2$, which is the weighted triangle inequality used below.

Finally, write $M = U \text{diag}(\xi_1, \dots, \xi_d)U^\top$ with $U^\top U = I$. The hypothesis $M \succeq \alpha I$ forces every $\xi_i \geq \alpha$. Setting $y = U^\top v$ gives $v^\top M^{-1}v = \sum_i y_i^2/\xi_i \leq \alpha^{-1}\|v\|_2^2$. $\square$

Proof of Lemma 2. Let $\alpha > 0$ be the smallest eigenvalue of A . The Euclidean case of Lemma 1 yields $Q(x) \geq (\alpha/2)\|x\|_2^2 - \|c\|_2\|x\|_2$. The right-hand side tends to $+\infty$ as $\|x\|_2 \rightarrow \infty$ and is bounded from below, so $\inf_{x \in \mathcal{X}} Q(x)$ is a finite real. Take a minimizing sequence $(x_j) \subseteq \mathcal{X}$. The same growth bound keeps the sequence bounded, so a subsequence $x_{j_\ell} \rightarrow x_*$ exists. Closedness gives $x_* \in \mathcal{X}$ and continuity gives $Q(x_*) = \inf_{x \in \mathcal{X}} Q(x)$.

If x_* is a minimizer, then for every $z \in \mathcal{X}$ and $\theta \in (0, 1]$ convexity puts $x_* + \theta(z - x_*)$ in $\mathcal{X}$, whence $0 \leq \theta^{-1}(Q(x_* + \theta(z - x_*)) - Q(x_*)) = \langle Ax_* + c, z - x_* \rangle + (\theta/2)\|z - x_*\|_A^2$. Sending $\theta \downarrow 0$ yields the first-order condition.

Conversely, if that condition holds, the quadratic expansion $Q(z) - Q(x_*) = \langle Ax_* + c, z - x_* \rangle + \frac{1}{2}\|z - x_*\|_A^2$ is at least $\frac{1}{2}\|z - x_*\|_A^2 \geq 0$. A second distinct minimizer z would force $\|z - x_*\|_A^2 > 0$ and therefore $Q(z) > Q(x_*)$, a contradiction. $\square$

Proof of Lemma 3. After dropping a z -independent constant the objective is positive definite with curvature M . Lemma 2 supplies a unique minimizer and the inequality $\langle M(p - h), b - p \rangle \geq 0$, or equivalently $\langle h - p, p - b \rangle_M \geq 0$. Expanding $\|h - b\|_M^2 = \|h - p\|_M^2 + \|p - b\|_M^2 + 2\langle h - p, p - b \rangle_M$ and using nonnegativity of the last term yields (11).

If p and b both lie in the ball of radius r , the weighted triangle inequality of Lemma 1 gives $\|p - b\|_M \leq 2r$. If $\|h\|_M \leq r$, then h is feasible and makes the nonnegative objective zero, so it is the unique minimizer. If $n = \|h\|_M > r$, the candidate $p = (r/n)h$ satisfies $\|p\|_M = r$. For an arbitrary feasible z ,

$$\langle h - p, z - p \rangle_M = (1 - r/n)(\langle h, z \rangle_M - (r/n)\|h\|_M^2).$$

Weighted Cauchy Schwarz and $\|z\|_M \leq r$ bound the second factor by $nr - (r/n)n^2 = 0$. The outer factor is nonnegative, so the first-order condition holds and p is the unique projection. $\square$

Proof of Lemma 4. First-order optimality of the perturbed objective at x , compared with z_* , gives $\langle Ax + c - r, z_* - x \rangle \geq 0$, hence $\langle Ax + c - r, x - z_* \rangle \leq 0$. First-order optimality of the unperturbed objective at z_* , compared with x , gives $\langle Az_* + c, x - z_* \rangle \geq 0$. Subtracting these two inner products produces $\|x - z_*\|_A^2 - \langle r, x - z_* \rangle \leq 0$. Lemma 1 then yields $\|x - z_*\|_A^2 \leq \|r\|_{A^{-1}}\|x - z_*\|_A$. The claim $\|x - z_*\|_A \leq \|r\|_{A^{-1}}$ is immediate if $x = z_*$, and follows by division if not.

The quadratic growth identity of Lemma 2 applied to the perturbed objective rearranges to $Q(x) - Q(z_*) \leq \langle r, x - z_* \rangle - \frac{1}{2}\|x - z_*\|_A^2$. Writing $s = \|x - z_*\|_A$ and $\rho = \|r\|_{A^{-1}}$, the right-hand side is at most $\rho s - \frac{1}{2}s^2 = \frac{1}{2}\rho^2 - \frac{1}{2}(s - \rho)^2 \leq \frac{1}{2}\rho^2$. The left-hand side is nonnegative because z_* minimizes Q . $\square$

Proof of Lemma 5. Let z_t minimize F_t . Rewrite $\Delta_t = [F_t(x_t) - m_t] - [F_{t-1}(x_t) - m_{t-1}]$. The second bracket is nonnegative, so $\Delta_t \leq F_t(x_t) - F_t(z_t)$. Lemma 4 then gives $\Delta_t \leq \frac{1}{2}\|r_t\|_{A_t^{-1}}^2$.

For the linear bound, $m_{t-1} \leq F_{t-1}(z_t)$ implies $\Delta_t \leq f_t(x_t) - f_t(z_t)$. Expanding the current quadratic at x_t and using $H_t \succeq 0$ produces $f_t(x_t) - f_t(z_t) \leq \langle g_t, x_t - z_t \rangle$. Lemma 1 and Lemma 4 finish the estimate $\Delta_t \leq \|g_t\|_{A_t^{-1}}\|r_t\|_{A_t^{-1}}$. Both bounds hold for the same Δ_t . $\square$

Proof of Lemma 6. Summing $\Delta_t = f_t(x_t) + m_{t-1} - m_t$ telescopes the optimal values and uses $m_0 = 0$, giving $\sum_t \Delta_t = \sum_t f_t(x_t) - m_T$. Substituting into the right-hand side of (13) produces $R_0(u) + \sum_t f_t(x_t) - F_T(u) = R_T(u)$. The inequality follows because $F_T(u) \geq m_T$. $\square$

Lemma 13 (Constants and first-round linear minimization). *Adding a decision-independent constant c_t to each quadratic loss leaves Δ_t , the instantaneous regret, the played points, the gradients and the curvature unchanged. If $\mathcal{X}$ is nonempty, compact and convex, every continuous convex quadratic on $\mathcal{X}$ attains its minimum, even if the curvature is only positive semidefinite. The least-Euclidean-norm point of the minimizer set exists and is unique. First-order optimality $\langle Bx + c, u - x \rangle \geq 0$ characterizes minimizers of $Q(x) = \frac{1}{2}x^\top Bx + c^\top x + C$ for $B \succeq 0$, without positive definiteness.*

Proof. Write $f'_t = f_t + c_t$. Then $F'_t = F_t + \sum_{s \leq t} c_s$ and $m'_t = m_t + \sum_{s \leq t} c_s$, so $\Delta'_t = f_t(x_t) + c_t + m_{t-1} + \sum_{s < t} c_s - m_t - \sum_{s \leq t} c_s = \Delta_t$, while $f'_t(x_t) - f'_t(u) = f_t(x_t) - f_t(u)$.

If $\mathcal{X}$ is compact, a minimizing sequence of a continuous convex quadratic has a convergent subsequence whose limit lies in $\mathcal{X}$ and attains the infimum. The minimizer set is nonempty, closed and contained in a compact set, hence compact, and it is convex. The Euclidean-norm-squared function has a unique minimizer on that set: if two distinct points both minimized it, their midpoint would remain feasible and strictly decrease the squared norm by the parallelogram identity.

Finally, if x minimizes Q with $B \succeq 0$, then for every feasible u and $\alpha \in (0, 1]$ convexity keeps $x + \alpha(u - x)$ feasible, so $0 \leq \alpha^{-1}(Q(x + \alpha(u - x)) - Q(x)) = \langle Bx + c, u - x \rangle + (\alpha/2)\|u - x\|_B^2$. Sending $\alpha \downarrow 0$ gives the first-order condition. Conversely, the quadratic expansion $Q(u) - Q(x) = \langle Bx + c, u - x \rangle + \frac{1}{2}\|u - x\|_B^2$ is nonnegative whenever that condition holds. Neither direction uses $B \succ 0$. The first round of the safeguarded update with $\lambda = 0$ invokes only this existence statement and never inverts the zero matrix. $\square$

Proof of Lemma 9. Expand n_t as a sum of indicators: $\sum_t n_t = \sum_t \sum_{s < t} \mathbf{1}\{\tau_s \geq t\}$. The finite sum may be reordered as $\sum_s \sum_{t=s+1}^T \mathbf{1}\{t \leq \tau_s\}$. For fixed s the inner sum counts the integers $s+1, \dots, \min\{\tau_s, T\}$, hence has cardinality $\min\{\tau_s - s, T - s\}$, with the convention that $\tau_s = \infty$ contributes $T - s$.

The bound $n_t \leq \sigma_T$ gives $M_{T,\lambda} \leq \sigma_T h_{T,\lambda}$. For the square-root bound, note that $\lambda \geq 0$ implies $M_{T,\lambda} \leq \sum_t n_t/t$. If $\mathcal{D}_T = 0$ every n_t vanishes. If $\mathcal{D}_T > 0$, set $a = \sqrt{\mathcal{D}_T} > 0$ and split $t \leq a$ from $t > a$. Since $\mathcal{U}_t \subseteq [t-1]$ one has $n_t \leq t$, so $\sum_{t \leq a} n_t/t \leq \#\{t \in [T] : t \leq a\} \leq a$. For $t > a$ one has $1/t < 1/a$, hence $\sum_{t > a} n_t/t \leq \mathcal{D}_T/a = \sqrt{\mathcal{D}_T}$. Adding the two pieces yields $M_{T,\lambda} \leq 2\sqrt{\mathcal{D}_T}$.

Finally $1/(t + \lambda) \leq 1/t$. For each integer $t \geq 2$ and $x \in [t-1, t]$ one has $1/x \geq 1/t$, so $1/t \leq \int_{t-1}^t x^{-1} dx$. Summing from $t = 2$ to T and adding the $t = 1$ term produces $h_{T,\lambda} \leq \sum_{t=1}^T 1/t \leq 1 + \int_1^T x^{-1} dx = 1 + \log T$, which also holds for $T = 1$. $\square$

B Known-quadratic specialization

The minimax upper bound of Theorem 6 and the first display of Theorem 4 use the known-Hessian algorithm RG-FTRL. We record it here because the lower-bound proofs rely on its guarantees. Losses are $f_t(x) = \frac{1}{2}x^\top Hx + a_t^\top x$ with a known $H \succ 0$ and $\|a_t\|_{H^{-1}} \leq L$. Hints $\hat{a}_t$ are issued before x_t and then frozen; write $e_t = a_t - \hat{a}_t$, $E_t = \sum_{s \in \mathcal{U}_t} e_s$, $b_t = \sum_{s \in \mathcal{V}_t} a_s$, $\hat{b}_t = \sum_{s \in \mathcal{V}_t} \hat{a}_s$, $\mathcal{C}_t = \{b : \|b\|_{H^{-1}} \leq k_t L\}$, $q_t = \arg \min_{b \in \mathcal{C}_t} \frac{1}{2}\|b - \hat{b}_t\|_{H^{-1}}^2$, $r_t = b_t - q_t$, $S_t^{\text{obs}} = \sum_{s \in \mathcal{O}_t} a_s$, and $A_t = (t + \lambda)H$. The update is $x_t = \arg \min_{x \in \mathcal{X}} \{\frac{1}{2}x^\top A_t x + \langle S_t^{\text{obs}} + q_t, x \rangle\}$. Define $Q_T^{\text{cur}} = \sum_t \|e_t\|_{A_t^{-1}}^2$, $Q_T^{\text{miss}} = \sum_t \|E_t\|_{A_t^{-1}}^2$, and $Q_T^{\text{joint}} = \sum_t \|E_t + e_t\|_{A_t^{-1}}^2$.

Theorem 7 (Perturbed quadratic potential). *Let $f_t(x) = \frac{1}{2}x^\top H_t x + a_t^\top x$ with $H_t \succeq 0$ and $R_0(x) = \frac{1}{2}x^\top \Lambda x$ with $\Lambda \succeq 0$. Assume $0 \in \mathcal{X}$ and $A_t = \Lambda + \sum_{s \leq t} H_s \succ 0$ for every $t \geq 1$. If the learner minimizes a computable positive-definite quadratic $\tilde{F}_t$ and $r_t = \nabla F_t(x_t) - \nabla \tilde{F}_t(x_t)$, then for every $u \in \mathcal{X}$*

$$R_T(u) \leq \frac{1}{2}u^\top \Lambda u + \sum_{t=1}^T \min \left\{ \frac{1}{2}\|r_t\|_{A_t^{-1}}^2, \|g_t\|_{A_t^{-1}}\|r_t\|_{A_t^{-1}} \right\} - [F_T(u) - m_T].$$

The terminal bracket may be dropped. The vectors r_t and the true A_t need not be available online.

Proof. Because x_t minimizes $\tilde{F}_t$, Lemma 2 gives $\langle \nabla \tilde{F}_t(x_t), z - x_t \rangle \geq 0$ for all $z \in \mathcal{X}$. The definition of r_t rewrites this as $\langle \nabla F_t(x_t) - r_t, z - x_t \rangle \geq 0$. The map $x \mapsto F_t(x) - r_t^\top x$ has curvature $A_t \succ 0$, so the sufficiency direction of Lemma 2 yields $x_t = \arg \min_{x \in \mathcal{X}} \{F_t(x) - r_t^\top x\}$. There is no circular definition: the algorithm first computes x_t from $\tilde{F}_t$, after which the analyst defines the numerical vector r_t at that already fixed point; in the perturbed program r_t no longer depends on the dummy variable x .

The regularizer R_0 is nonnegative and vanishes at the feasible point 0, so $m_0 = 0$. Each F_t for $t \geq 1$ is positive definite, so Lemma 2 supplies a minimizer. Lemma 5 therefore applies every round, and substituting the resulting bounds into Lemma 6 completes the proof. $\square$

Theorem 8 (Cancellation and robustness under a known Hessian). *Run RG-FTRL as above. Write $\rho_t = \|E_t + e_t\|_{H^{-1}}$. Then for every $u \in \mathcal{X}$,*

$$R_T(u) \leq \frac{\lambda}{2}\|u\|_H^2 + \sum_{t=1}^T \frac{1}{t + \lambda} \min \left\{ \frac{1}{2}\rho_t^2, 2L\rho_t, 4L^2k_t \right\}. \quad (25)$$

Moreover $\|x_t\|_H \leq tL/(t + \lambda) \leq L$, $\|g_t\|_{H^{-1}} \leq 2L$, and $\|r_t\|_{H^{-1}} \leq \min\{\|E_t + e_t\|_{H^{-1}}, 2k_tL\}$.

Proof. The set $\mathcal{V}_t$ has k_t indices, so the triangle inequality and the coefficient bound give $\|b_t\|_{H^{-1}} \leq k_tL$. Thus $b_t \in \mathcal{C}_t$, and the projection q_t lies in $\mathcal{C}_t$ as well.

Lemma 3 with $M = H^{-1}$, $h = \hat{b}_t$, $p = q_t$ and $b = b_t$ gives $\|b_t - q_t\|_{H^{-1}} \leq \|b_t - \hat{b}_t\|_{H^{-1}} = \|E_t + e_t\|_{H^{-1}}$. Both points lie in the ball of radius k_tL , so $\|b_t - q_t\|_{H^{-1}} \leq 2k_tL$. Combined, these are the claimed bound on $\|r_t\|_{H^{-1}}$.

Write $c_t = S_t^{\text{obs}} + q_t$. Since $|\mathcal{O}_t| + k_t = (t - 1 - n_t) + (n_t + 1) = t$, one has $\|c_t\|_{H^{-1}} \leq |\mathcal{O}_t|L + k_tL = tL$. First-order optimality of the update at the feasible point 0 yields $\langle (t + \lambda)Hx_t + c_t, -x_t \rangle \geq 0$, hence $(t + \lambda)\|x_t\|_H^2 \leq -\langle c_t, x_t \rangle \leq \|c_t\|_{H^{-1}}\|x_t\|_H \leq tL\|x_t\|_H$ by Lemma 1. If $x_t = 0$ the bound holds; otherwise division by the positive H -norm and by $t + \lambda$ gives $\|x_t\|_H \leq tL/(t + \lambda) \leq L$.

Symmetry and positive-definiteness of H give $\|Hx_t\|_{H^{-1}}^2 = x_t^\top Hx_t = \|x_t\|_H^2$, so $\|g_t\|_{H^{-1}} = \|Hx_t + a_t\|_{H^{-1}} \leq \|x_t\|_H + \|a_t\|_{H^{-1}} \leq 2L$.

The true cumulative objective is $F_t(x) = \frac{t+\lambda}{2}x^\top Hx + \langle \sum_{s \leq t} a_s, x \rangle$ and the algorithmic objective is $\tilde{F}_t(x) = \frac{t+\lambda}{2}x^\top Hx + \langle S_t^{\text{obs}} + q_t, x \rangle$. The partition $[t] = \mathcal{O}_t \cup \mathcal{V}_t$ yields $\sum_{s \leq t} a_s = S_t^{\text{obs}} + b_t$, so $F_t - \tilde{F}_t = \langle r_t, \cdot \rangle$ identically. The perturbation of Theorem 7 is exactly r_t ; there is no leftover curvature error.

Since $A_t = (t + \lambda)H$, one has $\|r_t\|_{A_t^{-1}}^2 = \|r_t\|_{H^{-1}}^2/(t + \lambda)$ and $\|g_t\|_{A_t^{-1}} = \|g_t\|_{H^{-1}}/\sqrt{t + \lambda}$. The quadratic branch of Lemma 5 and the projection bound give $\Delta_t \leq \rho_t^2/(2(t + \lambda))$. The linear branch

with $\|g_t\|_{H^{-1}} \leq 2L$ and $\|r_t\|_{H^{-1}} \leq \rho_t$ gives $\Delta_t \leq 2L\rho_t/(t + \lambda)$. The same linear branch with $\|r_t\|_{H^{-1}} \leq 2k_tL$ gives $\Delta_t \leq 4L^2k_t/(t + \lambda)$. The three bounds hold for the same Δ_t . Lemma 6, after dropping the nonpositive terminal term, produces (25). $\square$

Theorem 9 (Simultaneous residual and worst-case guarantees). *Under the conditions of Theorem 8, for every $\eta > 0$ and every $u \in \mathcal{X}$ one has simultaneously $R_T(u) \leq (\lambda/2)\|u\|_H^2 + \frac{1}{2}Q_T^{\text{joint}}$,*

$$R_T(u) \leq \frac{\lambda}{2}\|u\|_H^2 + \frac{1+\eta}{2}Q_T^{\text{cur}} + \frac{1+\eta^{-1}}{2}Q_T^{\text{miss}}, \quad (26)$$

and

$$R_T(u) \leq \frac{\lambda}{2}\|u\|_H^2 + 4L^2 \left[h_{T,\lambda} + \min\{\sigma_T h_{T,\lambda}, 2\sqrt{\mathcal{D}_T}\} \right]. \quad (27)$$

In particular, with $\eta = 1$,

$$R_T(u) \leq \frac{\lambda}{2}\|u\|_H^2 + \min\left\{Q_T^{\text{cur}} + Q_T^{\text{miss}}, 4L^2 \left[h_{T,\lambda} + \min\{\sigma_T h_{T,\lambda}, 2\sqrt{\mathcal{D}_T}\} \right] \right\}. \quad (28)$$

For $\lambda = 0$ the comparator term vanishes and the worst-case bound is $4L^2[(1 + \log T) + \min\{\sigma_T(1 + \log T), 2\sqrt{\mathcal{D}_T}\}]$.

Proof. Selecting the first per-round bound in (25) produces $R_T(u) \leq (\lambda/2)\|u\|_H^2 + \frac{1}{2}\sum_t \|E_t + e_t\|_{H^{-1}}^2/(t + \lambda)$, which is $(\lambda/2)\|u\|_H^2 + \frac{1}{2}Q_T^{\text{joint}}$ because $A_t^{-1} = H^{-1}/(t + \lambda)$. Applying (10) to $v = e_t$, $w = E_t$ and $M = A_t^{-1}$, summing, multiplying by 1/2 and substituting yields (26). The parameter η is used only in the analysis; it is not a learning rate.

Selecting the third per-round bound produces $R_T(u) \leq (\lambda/2)\|u\|_H^2 + 4L^2\sum_t k_t/(t + \lambda) = (\lambda/2)\|u\|_H^2 + 4L^2(h_{T,\lambda} + M_{T,\lambda})$. Lemma 9 upgrades this to (27). With $\eta = 1$ the two remaining bounds share the same regularizer term and may be minimized, giving (28). The last display uses $h_{T,\lambda} \leq 1 + \log T$ at $\lambda = 0$. $\square$

C Proofs for CG-RG-FTRL

Proof of Lemma 7. Specialize (7) at $x = x_t$ and rearrange: $f_t(x_t) - f_t(u) \leq -\langle g_t, u - x_t \rangle - \frac{1}{2}\|u - x_t\|_{P_t}^2 = \ell_t(x_t) - \ell_t(u)$, using $\ell_t(x_t) = 0$. Summing yields the regret transfer.

By definition x_t minimizes Q_t^{alg} in (3). Define $\tilde{F}_t(x) = Q_t^{\text{alg}}(x) + \frac{1}{2}\|x - x_t\|_{P_t}^2$. For every $x \in \mathcal{X}$ the first summand is at least $Q_t^{\text{alg}}(x_t)$ and the second is nonnegative and vanishes at x_t , so $\tilde{F}_t(x) \geq Q_t^{\text{alg}}(x_t) = \tilde{F}_t(x_t)$: the already computed x_t also minimizes $\tilde{F}_t$. This is an analytical function defined after the decision, not an implicit program whose center is unknown to the learner.

Differentiating term by term,

$$\begin{aligned} \nabla F_t(x) &= \lambda\mu x + \sum_{s \leq t} P_s(x - x_s) + \sum_{s \leq t} g_s, \\ \nabla \tilde{F}_t(x) &= \lambda\mu x + \sum_{s \leq t} P_s(x - x_s) + \sum_{s \in \mathcal{O}_t} g_s + q_t. \end{aligned}$$

Because $\mathcal{O}_t$ and $\mathcal{V}_t$ partition $[t]$, the difference is the constant vector $b_t^g - q_t = r_t^g$. Thus $F_t - \langle r_t^g, \cdot \rangle$ and $\tilde{F}_t$ differ by a constant and share the same minimizer. Both Hessians equal $\lambda\mu I + \sum_{s \leq t} P_s = A_t$. Substituting $x = x_t$ gives $\nabla \ell_t(x_t) = g_t$. Constants are absorbed by Lemma 13 and do not affect the potential or the transfer. $\square$

Proof of Lemma 8. The triangle inequality and Assumption 2 give $\|b_t^g\|_2 \leq \sum_{s \in \mathcal{V}_t} \|g_s\|_2 \leq k_t G$, so b_t^g lies in the projection ball of (1). Lemma 3 in the metric A_t^{-1} , applied to the point h_t^g and the feasible point b_t^g , yields $\|b_t^g - q_t\|_{A_t^{-1}} \leq \|b_t^g - h_t^g\|_{A_t^{-1}} = \|\sum_{s \in \mathcal{V}_t} (g_s - \hat{g}_s)\|_{A_t^{-1}} = \rho_t^g$. Feasibility of the ball gives $\|q_t\|_2 \leq k_t G$, hence $\|r_t^g\|_2 \leq \|b_t^g\|_2 + \|q_t\|_2 \leq 2k_t G$. Summing $P_s \succeq \mu I$ produces $A_t \succeq \mu(t + \lambda)I$. The inverse bound of Lemma 1 then yields $\|g_t\|_{A_t^{-1}}^2 \leq \|g_t\|_2^2 / (\mu(t + \lambda)) \leq G^2 / (\mu(t + \lambda))$. $\square$

Proof of Theorem 1. For $t \geq 1$ the surrogate F_t has curvature $A_t \succ 0$, so Lemma 2 supplies m_t . The regularizer $R_0(x) = \lambda\mu\|x\|_2^2/2$ is nonnegative and vanishes at the feasible point 0, so $m_0 = 0$.

Define the surrogate gap $\Delta_t^\ell = \ell_t(x_t) + m_{t-1} - m_t$. Lemma 7 places x_t as a minimizer of $F_t - \langle r_t^g, \cdot \rangle$ with current surrogate gradient g_t . The two already proved inequalities of Lemma 5 therefore give

$$\Delta_t^\ell \leq \frac{1}{2} \|r_t^g\|_{A_t^{-1}}^2, \quad \Delta_t^\ell \leq \|g_t\|_{A_t^{-1}} \|r_t^g\|_{A_t^{-1}}. \quad (29)$$

The first, with the first claim of Lemma 8, produces $(\rho_t^g)^2/2$. The second, with the first and third claims of Lemma 8, produces $G\rho_t^g/\sqrt{\mu(t + \lambda)}$. The inverse-norm bound together with the second claim of Lemma 8 yields $\|r_t^g\|_{A_t^{-1}} \leq \|r_t^g\|_2/\sqrt{\mu(t + \lambda)} \leq 2k_t G/\sqrt{\mu(t + \lambda)}$. Substituting this and $\|g_t\|_{A_t^{-1}} \leq G/\sqrt{\mu(t + \lambda)}$ into (29) produces the third branch $\Delta_t^\ell \leq 2G^2 k_t / (\mu(t + \lambda))$. All three inequalities control the same gap, so the per-round minimum is legitimate.

Lemma 6 applied to the surrogates gives $\sum_t [\ell_t(x_t) - \ell_t(u)] = (\lambda\mu/2)\|u\|_2^2 + \sum_t \Delta_t^\ell - [F_T(u) - m_T]$. The bracket is nonnegative. Dropping it and invoking the transfer of Lemma 7 produces (14).

Young's inequality (10) at $\eta = 1$ yields $\frac{1}{2}\|e_t^g + E_t^g\|_{A_t^{-1}}^2 \leq \|e_t^g\|_{A_t^{-1}}^2 + \|E_t^g\|_{A_t^{-1}}^2$. Selecting the first branch of (14) and summing produces the residual certificate of (15). Selecting the third branch and using $k_t = 1 + n_t$ produces $\sum_t 2G^2 k_t / (\mu(t + \lambda)) = (2G^2/\mu)(h_{T,\lambda} + M_{T,\lambda})$. The two certificates share the same regularizer and may be minimized. $\square$

Proof of Corollary 1. If every hint is exact then $e_t^g = E_t^g = 0$, so the first branch of (14) vanishes, as does the regularizer when $\lambda = 0$. If every hint is zero, then $h_t^g = 0$ lies in the projection ball; the nonnegative projection objective vanishes at $q = 0$, so $q_t = 0$. Thus $r_t^g = b_t^g$ and the diameter bound is unnecessary: $\|r_t^g\|_2 \leq k_t G$. Substituting into the linear branch of (29) produces $G^2 k_t / (\mu(t + \lambda))$, and summing improves the leading constant from 2 to 1. The third claim is $\|E_t^g\|_{A_t^{-1}}^2 = 0$ on those rounds. $\square$

D Proofs for the joint-delay envelope

Proof of Theorem 2. For the upper bound, any feasible backlog satisfies $0 \leq n_t \leq w_t$ and $\sum_t n_t \leq \delta$, the latter because the budget caps the sum by D and the capacities cap it by C_T . If $m = T$, then $n_t/t \leq w_t/t$ termwise and the first line of (16) is an upper bound.

If $m < T$, define the numerical candidate $v_t = w_t$ for $t \leq m$, $v_{m+1} = b$, and $v_t = 0$ for $t \geq m + 2$. Its sum is $C_m + b = \delta$. Writing $a = 1/(m + 1)$,

$$\sum_{t=1}^T \frac{n_t - v_t}{t} = \sum_{t=1}^m (n_t - w_t) \left(\frac{1}{t} - a \right) + \sum_{t=m+2}^T n_t \left(\frac{1}{t} - a \right) + a \left(\sum_{t=1}^T n_t - \delta \right). \quad (30)$$

The identity is an expansion: the coefficients of $-a$ on all differences are cancelled by the last term, and the $t = m + 1$ difference is left with coefficient a . In the first sum $n_t - w_t \leq 0$ and $1/t - a > 0$; in the second $n_t \geq 0$ and $1/t - a < 0$; in the last the sum-gap is nonpositive and $a > 0$. All three contributions are nonpositive, so $\sum_t n_t/t \leq \sum_t v_t/t$, which is the second line of (16).

To show attainability by a legal schedule, suppose that $m = T$. Withhold the first $\bar{\sigma}$ observations until the end of round T and return every other observation immediately. Before the decision at time t the missing set is exactly the first $\min\{\bar{\sigma}, t - 1\}$ indices, so $n_t = w_t$. The total backlog is $C_T = \delta \leq D$. The case $\bar{\sigma} = 0$ is instantaneous feedback.

If $m < T$, withhold the already produced portion of the first $\bar{\sigma}$ observations through the end of round m and return everything else immediately, so $n_t = w_t$ for $t \leq m$. At the end of round m the outstanding set has size $w_m + 1$. Because $w_{m+1} = \min\{\bar{\sigma}, m\} \leq w_m + 1$ and $b < w_{m+1}$, one may retain exactly b of those $w_m + 1$ items and return the rest, giving $n_{m+1} = b$. At the end of round $m + 1$ return the remaining b items together with the current observation; thereafter return everything immediately, so $n_t = 0$ for $t \geq m + 2$. The resulting backlog is v_t , its peak is at most $\bar{\sigma}$, and its sum is $\delta \leq D$. Matching upper and lower bounds complete the proof. $\square$

Proof of Lemma 10. Suppose round t misses the distinct indices $s_1, \dots, s_r < t$. Each index s_i contributes to the backlog at the decision rounds $s_i + 1, \dots, t$, hence $t - s_i$ times. Distinct indices are counted separately, so $\mathcal{D}_T \geq \sum_{i=1}^r (t - s_i)$. The ages $t - s_i$ are distinct positive integers; sorted increasingly, the i -th is at least i , and their sum is at least $1 + \dots + r = r(r + 1)/2$. (The closed form follows by adding the sum to its reversal: r pairs each equal $r + 1$.) Solving $r^2 + r - 2D \leq 0$ for an integer r yields the displayed bound on the peak. $\square$

Proof of Theorem 3. Lemma 9 gives $\sum_t n_t/t \leq 2\sqrt{\mathcal{D}_T} \leq 2\sqrt{D}$ on every legal schedule, hence on the maximizing one. A zero budget or a zero peak forces every $n_t = 0$.

Write $s = \bar{\sigma} \geq 1$. The capacity bound produces

$$\begin{aligned} \mathfrak{M}_T(s, D) &\leq \sum_{t=1}^{s+1} \frac{t-1}{t} + s \sum_{t=s+2}^T \frac{1}{t} = s + 1 - h_{s+1} + s(h_T - h_{s+1}) \\ &\leq s + s \log \frac{T}{s+1} \leq s[1 + \log(T/s)], \end{aligned}$$

using $h_{s+1} \geq 1$ and, for integers $b > a \geq 1$, the integral comparison $\sum_{t=a+1}^b t^{-1} \leq \int_a^b x^{-1} dx = \log(b/a)$. Empty sums vanish, so the bound also covers $T = s + 1$.

For the last claim use the m, b of Theorem 2. If $m \leq s$, every nonzero candidate lies in $t \leq s + 1$ and is at most the capacity, so $\mathfrak{M}_T(s, D) \leq \sum_{t=1}^{s+1} (t-1)/t \leq s \leq s[1 + \log(1 + D/s^2)]$. If $m \geq s + 1$, inflate a possible partial fill to a full capacity slot: $\mathfrak{M}_T(s, D) \leq s + s \log((m+1)/(s+1))$. When $m < T$ this comes from completing the sum through $m + 1$; when $m = T$ adding one more positive term is still an upper bound, so the same expression applies.

The prefix capacity is $C_m = \sum_{t=1}^{s+1} (t-1) + \sum_{t=s+2}^m s = s(s+1)/2 + s(m-s-1) = sm - s(s+1)/2$. The constraint $C_m \leq \delta \leq D$ rearranges to $(m+1)/(s+1) \leq D/(s(s+1)) + (s+3)/(2(s+1)) \leq D/s^2 + 1$, using $s(s+1) \geq s^2$ and $s+3 \leq 2s+2$ (i.e., $s \geq 1$). Substituting into the increasing logarithm finishes the proof. $\square$

Proof of Theorem 4. The realized schedule belongs to the feasible class of (4) with budgets $(\sigma_T, \mathcal{D}_T)$, so $M_T = \sum_t n_t/t \leq \mathfrak{M}_T(\sigma_T, \mathcal{D}_T)$. The proof of Theorem 1 already gives $R_T(u) \leq (2G^2/\mu)(h_T + M_T)$ on the third branch before any crude counting bound is applied, and the residual branch is $\sum_t \|e_t^g\|_{A_t^{-1}}^2 + \sum_t \|E_t^g\|_{A_t^{-1}}^2$. Substituting the comparison produces (18). The companion statement for RG-FTRL is the same replacement in (28) at $\lambda = 0$, using the identity $R_T(u) \leq 4L^2(h_T + M_T)$ already obtained in the proof of Theorem 9 before Lemma 9 is invoked. Both substitutions are analytical; the algorithms' inputs are unchanged. $\square$

Proof of Corollary 2. For all sufficiently large T one has $T/\log^2 T \geq 2$, hence $T/(2\log^2 T) \leq s_T \leq T/\log^2 T$. Integral comparison gives $\log(T+1) \leq h_T \leq 1 + \log T$, so $s_T h_T = \Theta(T/\log T)$. Since $s_T/T \rightarrow 0$, one has $s_T + 1 \leq T$ for large T and therefore $\frac{1}{2}s_T T \leq D_T \leq s_T T$. Taking square roots and using both sides of the bound on s_T yields $\sqrt{D_T} = \Theta(T/\log T)$.

The budget fills every capacity, so (16) gives exactly $\mathfrak{M}_T = s_T + 1 - h_{s_T+1} + s_T(h_T - h_{s_T+1})$. Piecewise integration produces $\log((T+1)/(s_T+2)) \leq h_T - h_{s_T+1} \leq \log(T/(s_T+1))$. With $s_T = \Theta(T/\log^2 T)$ both logarithms equal $2\log \log T + O(1)$, so the tail is $\Theta(s_T \log \log T)$ while the prefix is nonnegative and at most s_T . Thus $\mathfrak{M}_T = \Theta(T \log \log T / \log^2 T)$. Moreover $h_T/(T \log \log T / \log^2 T) \rightarrow 0$: the numerator times $\log^2 T$ grows at most like $\log^3 T$, while the denominator contains $T \log \log T$. Writing $y = \log T$, the elementary bound $e^y \geq y^{k+1}/(k+1)!$ shows that the exponential dominates every fixed power, so adjoining h_T does not change the stated order. $\square$

E Proofs of the information lower bounds

Proof of Lemma 11. For nonnegative integers r, s write $I_{r,s} = \int_0^1 p^r(1-p)^s dp$. If $r = 0$ then $I_{0,s} = 1/(s+1)$. If $r \geq 1$, integration by parts gives $I_{r,s} = [-p^r(1-p)^{s+1}/(s+1)]_0^1 + (r/(s+1))I_{r-1,s+1}$, the boundary terms vanishing. Iterating r times produces $I_{r,s} = r!s!/(r+s+1)!$, which also holds at $r = 0$ with the convention $0! = 1$.

For a concrete sequence with k pluses the likelihood given p is $p^k(1-p)^{n-k}$. The prior density is 1, so $\mathbb{E}[p \mid Z_1, \dots, Z_n] = I_{k+1,n-k}/I_{k,n-k} = (k+1)/(n+2)$. Doubling and subtracting 1 yields the first identity in (20); the $n = 0$ case is 0, matching the prior mean $1/2$ of p .

There are $\binom{n}{k}$ sequences with k pluses, so $\mathbb{P}(K_n = k) = \binom{n}{k} I_{k,n-k} = 1/(n+1)$ for $k = 0, \dots, n$: thus K_n is uniform on those integers. The elementary identities $\sum_{k=0}^n k = n(n+1)/2$ and $\sum_{k=0}^n k^2 = n(n+1)(2n+1)/6$ hold by induction (both sides vanish at $n = 0$; the increments when n is replaced by $n+1$ are $n+1$ and $(n+1)^2$ respectively). Therefore $\mathbb{E}[K_n] = n/2$ and $\mathbb{E}[K_n^2] = n(2n+1)/6$, whence $\mathbb{E}[(2K_n - n)^2] = n(n+2)/3$. Dividing by $(n+2)^2$ produces the second identity in (20).

Direct integration gives $\mathbb{E}[\theta^2] = \int_0^1 (2p-1)^2 dp = 1/3$. Conditionally on p one has $\mathbb{E}[Z_i \mid p] = \theta$, and for $i \neq j$ conditional independence yields $\mathbb{E}[Z_i Z_j \mid p] = \theta^2$, hence $\mathbb{E}[Z_i Z_j] = 1/3$. Finally $Z_i^2 = 1$ identically. $\square$

Proof of Theorem 5. Before block j begins, every observation from the first $j-1$ blocks has arrived. Given past actions and μ , each returned gradient recovers its coefficient, so the learner knows $Z_1, \dots, Z_{j-1}$. Repeated coefficients inside the current block add no independent sample. Until the end of block j , no observation containing Z_j has arrived.

The learner's internal randomness is independent of $(p, Z_1, \dots, Z_N)$. Conditionally on the past signs and the seed, every action in block j is a function of already available information and cannot create information about p or Z_j . The conditional mean of the current coefficient is therefore $\mathbb{E}[a_t \mid \text{available information}] = \varepsilon m_{j-1}$, where m_{j-1} is the posterior mean of Lemma 11. The conditional law is unaffected by deterministic or independently randomized post-processing of the history.

For any $x \in \mathcal{X}$ chosen given the available information,

$$\begin{aligned} \mathbb{E}[f_t(x) \mid \text{available information}] &= \frac{\mu}{2}x^2 + \varepsilon m_{j-1}x \\ &= \frac{\mu}{2}\left(x + \varepsilon m_{j-1}/\mu\right)^2 - \frac{\varepsilon^2 m_{j-1}^2}{2\mu} \\ &\geq -\frac{\varepsilon^2 m_{j-1}^2}{2\mu}. \end{aligned} \tag{31}$$

Since m_{j-1} is a conditional mean of a $[-1, 1]$ -valued random variable, $|m_{j-1}| \leq 1$, so $x = -\varepsilon m_{j-1}/\mu$ is feasible and attains equality. Randomization cannot improve the value, because the quadratic term is nonnegative for every realized x .

Taking the unconditional expectation, using (20), and summing over blocks, every learner's expected cumulative loss is at least

$$-\frac{\varepsilon^2}{2\mu} \sum_{j=1}^N w_j \frac{j-1}{3(j+1)}. \tag{32}$$

The legal algorithm that plays $-\varepsilon m_{j-1}/\mu$ throughout block j attains every equality.

For the offline comparator, write $S = \sum_j w_j Z_j$. The cumulative loss $\mu T x^2/2 + \varepsilon S x$ is minimized at $u^* = -\varepsilon S/(\mu T)$, with value $-\varepsilon^2 S^2/(2\mu T)$. Feasibility holds because $|S| \leq \sum_j w_j = T$. Lemma 11 gives

$$\begin{aligned} \mathbb{E}[S^2] &= \sum_j w_j^2 \mathbb{E}[Z_j^2] + 2 \sum_{i < j} w_i w_j \mathbb{E}[Z_i Z_j] \\ &= \sum_j w_j^2 + \frac{2}{3} \sum_{i < j} w_i w_j = \frac{1}{3} T^2 + \frac{2}{3} \sum_j w_j^2, \end{aligned}$$

where $T^2 = \sum_j w_j^2 + 2 \sum_{i < j} w_i w_j$. The comparator's expected cumulative loss is therefore $-(\varepsilon^2/(2\mu)) [T/3 + (2/(3T)) \sum_j w_j^2]$.

Subtracting this value from (32) and using $(j-1)/(j+1) = 1 - 2/(j+1)$ together with $\sum_j w_j = T$ produces

$$\begin{aligned} \mathbb{E}[R_T(u^*)] &\geq \frac{\varepsilon^2}{2\mu} \left[\frac{T}{3} + \frac{2}{3T} \sum_j w_j^2 - \frac{1}{3} \sum_j w_j + \frac{2}{3} \sum_j \frac{w_j}{j+1} \right] \\ &= \frac{\varepsilon^2}{3\mu} \left[\frac{\sum_j w_j^2}{T} + \sum_j \frac{w_j}{j+1} \right]. \end{aligned}$$

The posterior-mean algorithm attains (32) with equality, and the comparator loss does not depend on the algorithm's actions, so equality is attained in the Bayes risk. $\square$

Proof of Corollary 3. Substitute $w_j = \ell$ into (21): $\sum_j w_j^2/T = N\ell^2/(N\ell) = \ell$ and $\sum_j w_j/(j+1) = \ell(h_{N+1} - 1)$, which add to the stated Bayes value. Inside a block the i -th decision misses $i-1$ past observations, $i = 1, \dots, \ell$, so the peak is $\ell-1$; each block contributes $\sum_{i=1}^{\ell} (i-1) = \ell(\ell-1)/2$ to the total backlog.

The expected regret of any fixed algorithm, averaged against the prior over the finitely many sign sequences, is at least the Bayes risk. Hence at least one sign sequence has algorithmic expected regret no smaller than that average: otherwise every term would be strictly smaller and a positive-weight average would remain strictly smaller, a contradiction. That sequence together with the fixed block schedule is the required deterministic instance.

For the upper bound, fix an arbitrary sign sequence. In the known-quadratic model one has $H = \mu$ and $L = \varepsilon/\sqrt{\mu}$. Zero hints remain zero after the aggregate projection. On the i -th round of block b the outstanding coefficient sum is $i\varepsilon Z_b$, so $\rho_t = i\varepsilon/\sqrt{\mu}$ at $t = (b-1)\ell + i$. The per-round certificate of Theorem 8 is therefore $(\varepsilon^2/\mu) \cdot \min\{i^2/2, 2i, 4i\}/((b-1)\ell + i) = (\varepsilon^2/\mu) \cdot \min\{i^2/2, 2i\}/((b-1)\ell + i)$. For $1 \leq i \leq \ell$ one has $(b-1)\ell + i - bi = (b-1)(\ell-i) \geq 0$, hence $i/((b-1)\ell + i) \leq 1/b$ and the certificate is at most $2\varepsilon^2/(\mu b)$. Summing inside the block and then over $b = 1, \dots, N$ produces $(2\varepsilon^2\ell/\mu)h_N$. Since $h_N \leq h_{N+1}$, the ratio of the upper certificate to the Bayes lower bound is at most 6. $\square$

Proof of Lemma 12. Because $D \geq 1$ the integer part of the radical is at least 1; together with $\bar{\sigma} \geq 1$ and $T-1 \geq 1$ this forces $k \geq 1$. The relation $k \leq T-1$ gives $\ell \leq T$. The radical bound is equivalent to $k(k+1)/2 \leq D$, so $C \leq D$. Both floor expressions are at least 1, hence $N \geq 1$.

Lemma 10 shows that every feasible schedule under the original budgets has peak at most k . The original feasible class therefore coincides with the class whose peak cap is k , and $\mathfrak{M}_T(\bar{\sigma}, D) = \mathfrak{M}_T(k, D)$. By definition N equals at least one of the two floors in its formula.

If $N = \lfloor T/\ell \rfloor$, then $T < (N+1)\ell$, so $T/k < (\ell/k)(N+1) \leq 2(N+1)$ because $\ell/k = 1 + 1/k \leq 2$. The time majorant of Theorem 3 gives $\mathfrak{M}_T(k, D) \leq k[1 + \log(T/k)] \leq k[1 + \log 2 + \log(N+1)]$. If $N \neq \lfloor T/\ell \rfloor$, then necessarily $N = \lfloor D/C \rfloor$, so $D < (N+1)C$ and $D/k^2 < ((k+1)/(2k))(N+1) \leq N+1$. The budget majorant of Theorem 3 gives $\mathfrak{M}_T(k, D) \leq k[1 + \log(1 + D/k^2)] \leq k[1 + \log(N+2)] \leq k[1 + \log 2 + \log(N+1)]$, using $N+2 \leq 2(N+1)$. Equality of the two floors is already covered by the first case.

Since $N \geq 1$ one has $h_N \geq 1$. Because $1/x$ decreases, $h_N = \sum_{j=1}^N j^{-1} \geq \sum_{j=1}^N \int_j^{j+1} x^{-1} dx = \log(N+1)$. The common majorant is therefore at most $(2 + \log 2)kh_N < 3\ell h_N$, using $\log 2 < 1$ (since $\int_1^2 x^{-1} dx < \int_1^2 1 dx = 1$) and $k < \ell$.

Finally $N\ell \leq T$ makes the block partition feasible. Each length- ℓ block consumes $\ell(\ell-1)/2 = C$ total backlog and has peak $\ell-1 = k$; each singleton consumes none. The total backlog is $NC \leq D$ and the peak is $k \leq \bar{\sigma}$. $\square$

Proof of Theorem 6. For the upper bound, Run RG-FTRL with $H = \mu$, $L = \varepsilon/\sqrt{\mu}$, $\lambda = 0$ and every hint equal to zero. Then $q_t = 0$ and $\|r_t\|_{H^{-1}} = \|\sum_{s \in \mathcal{V}_t} a_s\|_{H^{-1}} \leq k_t L$. Theorem 8 already gives $\|g_t\|_{H^{-1}} \leq 2L$. The linear branch of Lemma 5 produces $\Delta_t \leq 2L^2 k_t/t$. The regularizer vanishes, and the potential telescope with a nonnegative terminal term yields $R_T(u^*) \leq 2L^2 \sum_t (1 + n_t)/t = (2\varepsilon^2/\mu)(h_T + M_T) \leq (2\varepsilon^2/\mu)[h_T + \mathfrak{M}_T(\bar{\sigma}, D)]$. The bound holds on every instance of the class, so taking the supremum over instances and the infimum over algorithms produces the upper bound of (22).

For the undelayed lower bound, specialize Theorem 5 to $N = T$ and $w_j = 1$ for all j . Every observation returns immediately, which is feasible for every nonnegative delay budget. Then $\mathcal{B}(1, \dots, 1) = (\varepsilon^2/(3\mu))[1 + (h_{T+1} - 1)] = (\varepsilon^2/(3\mu))h_{T+1} \geq (\varepsilon^2/(3\mu))h_T$. For any randomized algorithm the prior-average regret is at least this value. Viewing the algorithmic expectation as a real number on each deterministic instance, a finite weighted average cannot exceed the maximum, so a budget-feasible deterministic instance attains at least that value. The argument applies to every algorithm, hence $\mathcal{R}_T^*(\bar{\sigma}, D) \geq (\varepsilon^2/(3\mu))h_T$. If $\mathfrak{M}_T = 0$ this already exceeds the target lower bound and the proof stops.

For the delayed lower bound, suppose $\mathfrak{M}_T > 0$, which forces $T \geq 2$, $\bar{\sigma} \geq 1$ and $D \geq 1$. Using the (k, ℓ, C, N) of Lemma 12, form N leading blocks of length ℓ and fill the remainder with singletons. The schedule meets the original budgets. All terms in (21) are nonnegative; retaining only the second part of the first N blocks gives $\mathcal{B} \geq (\varepsilon^2\ell/(3\mu)) \sum_{j=1}^N 1/(j+1)$. For $j \geq 1$ one has $j+1 \leq 2j$, so $1/(j+1) \geq 1/(2j)$ and $\mathcal{B} \geq (\varepsilon^2\ell/(6\mu))h_N \geq (\varepsilon^2/(18\mu))\mathfrak{M}_T(\bar{\sigma}, D)$, where the final inequality uses Lemma 12. Finite averaging versus the maximum again produces, for every algorithm, a budget-feasible deterministic instance attaining this lower bound.

For each algorithm the class worst case is at least both the undelayed and the delayed lower bounds, even though the two bad instances may differ. Thus $\mathcal{R}_T^*(\bar{\sigma}, D) \geq (\varepsilon^2/\mu) \max\{h_T/3, \mathfrak{M}_T/18\}$. For nonnegative a, b one has $\max\{a, b\} \geq (a+b)/2$ because $a+b \leq 2\max\{a, b\}$. Therefore $\max\{h_T/3, \mathfrak{M}_T/18\} \geq h_T/6 + \mathfrak{M}_T/36 \geq (h_T + \mathfrak{M}_T)/36$. This includes the degenerate case already treated separately. $\square$

Proof of Corollary 4. For the upper bound, the argument of Theorem 6 used only $\|r_t\|_{H^{-1}} \leq k_t L$, $\|g_t\|_{H^{-1}} \leq 2L$ and the potential identity, all of which Theorem 8 establishes for an arbitrary known H and an arbitrary closed convex domain. The zero-hint algorithm therefore satisfies $R_T(u) \leq 2L^2(h_T + M_T)$ for every $u \in \mathcal{X}$. The cumulative quadratic is coercive and strictly convex, so a best static point exists; evaluating at that point and using $M_T \leq \mathfrak{M}_T$ on feasible schedules produces the upper bound.

For the lower bound, choose v with $v^\top H v = 1$ (normalize an arbitrary nonzero vector). On the sign blocks of Lemma 11 set $a_t = L Z_j H v$. Then $\|a_t\|_{H^{-1}}^2 = L^2 v^\top H H^{-1} H v = L^2$. Any action decomposes uniquely as $x_t = \xi_t v + z_t$ with $\xi_t = v^\top H x_t$ and $v^\top H z_t = 0$. Inside block j , $f_t(x_t) = \frac{1}{2}\xi_t^2 + L Z_j \xi_t + \frac{1}{2}\|z_t\|_H^2$. Until feedback arrives, the conditional mean of the current sign remains m_{j-1} ; returned gradients minus the known $H x_s$ reveal only the corresponding past sign and add no information about the current block. The conditional expectation therefore satisfies

$$\begin{aligned} \mathbb{E}[f_t(x_t) \mid \text{available information}] &= \frac{1}{2}(\xi_t + L m_{j-1})^2 - L^2 m_{j-1}^2/2 + \frac{1}{2}\|z_t\|_H^2 \\ &\geq -L^2 m_{j-1}^2/2. \end{aligned}$$

The inequality holds for every real ξ_t and does not require truncating actions on an unbounded domain. If the algorithm incurs infinite expected quadratic cost the lower bound holds a fortiori; otherwise expectations may be taken termwise.

Write $S = \sum_j w_j Z_j$. The unconstrained minimizer of the cumulative objective is $u^* = -(LS/T)v$, whose H -norm is at most L and therefore lies in $\mathcal{X}$, with value $-L^2 S^2/(2T)$. Subtracting the conditional-loss bound, $\mathbb{E}[m_{j-1}^2]$ from Lemma 11, and $\mathbb{E}[S^2]$ from the computation in the proof of Theorem 5 produces $\mathbb{E}[R_T(u^*)] \geq (L^2/3)[(\sum_j w_j^2)/T + \sum_j w_j/(j+1)]$, which is the block lower bound of Theorem 5 with L^2 in place of ε^2/μ . Singleton blocks give the lower bound $L^2 h_T/3$; the

budgeted blocks of Lemma 12 give $L^2\mathfrak{M}_T/18$. Finite averaging versus the maximum converts both into deterministic worst cases for every algorithm; the numerical comparison already proved at the end of Theorem 6 combines them into $L^2(h_T + \mathfrak{M}_T)/36$. $\square$

Proof of Corollary 5. For the upper bound, the domain is nonempty, compact and convex, and contains the origin. The certificate $P_t = \mu I$ satisfies Assumption 2 by strong convexity; the uniform gradient bound is Assumption 2. Run the safeguarded update with $\lambda = 0$ and zero hints. Corollary 1 improves the safeguard constant from 2 to 1, so $R_T(u) \leq (G^2/\mu)(h_T + M_T) \leq (G^2/\mu)[h_T + \mathfrak{M}_T(\bar{\sigma}, D)]$. Cumulative loss is continuous and the domain is compact, so a best comparator exists; taking the class supremum produces the upper bound.

For the lower bound, set $H = \mu I$ and $L = G/(2\sqrt{\mu})$. The H -ball of radius L is exactly $\mathcal{X}$, because $\|x\|_H = \sqrt{\mu}\|x\|_2$. The quadratic subclass $f_t(x) = (\mu/2)\|x\|_2^2 + a_t^\top x$ with $\|a_t\|_2 \leq G/2$ is μ -strongly convex, and for every feasible x $\|\nabla f_t(x)\|_2 = \|\mu x + a_t\|_2 \leq \mu\|x\|_2 + \|a_t\|_2 \leq \mu \cdot G/(2\mu) + G/2 = G$. It is therefore a legal subclass of $\mathcal{F}_{\mu,G}$, and $\|a_t\|_{H^{-1}} = \|a_t\|_2/\sqrt{\mu} \leq L$. Corollary 4 supplies the all-algorithm lower bound $L^2(h_T + \mathfrak{M}_T)/36 = G^2(h_T + \mathfrak{M}_T)/(144\mu)$ on this subclass. The worst case of any algorithm over the full class is at least its worst case over the subclass; taking the infimum over algorithms preserves the inequality. $\square$